

WEB API Manual

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Seylan – API Integration with BLICQ (Pvt) Ltd

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Change Log

Version	Date	Remarks

Integrating with Bank APIs

All APIs available in this document are exposed as REST APIs with JSON data format. In order to invoke an API, client should have registered with the bank and obtained a valid API Key.

Bank Service Endpoint

```
Host      : dev.apigateway1.seylan.ad / dev.apigateway1.seylan.lk  
             (indicated as <<sbk-gateway>> in all sample URLs)  
Port      : 2260/2270/2250  
Accept    : application/json
```

Sample Request URL

Sample GET request would be as follows:

```
https://<<sbk-gateway:port>>/Inquiries/1.0/SampleService?AccountCategory=EXT&Account  
Number=008000102000001
```

Sample Response

Sample GET Response would be as follows:

```
{  
  "Sample_Service": {  
    "Status": {  
      "Transaction_Reference": "E27106142618$$AA00XXX4662204",  
      "Timestamp": "2020-08-06T08:56:23.132Z",  
      "Code": "0000"  
    },  
    "Sample_Information": {  
      "Account_branch": "0180",  
      "Basic_part_of_account_number": "109810",  
      "Account_suffix": "011",  
      "Account_short_name": "NEO PLC",  
      "Arabic_short_name": "NEO PLC",  
      "Customer_mnemonic": "109810",  
      "Customer_location": "ABA",  
      "Customer_full_name": "NEO PLC",  
      "Account_type": "CA",  
      "Customer_type": "CA",  
    }  
  }  
}
```

Similar to above, every response will consist of Status section. Based on availability, following information will be returned.

- Transaction Reference : Unique reference number allocated to this API call
- Timestamp : Timestamp of API response
- Code : Status code returned from API call. A successful API call will carry status code '0000'.
- Message : If applicable, Error message
- Description : If applicable, Error description

Legends

Input Format/Data Type

Value	Description	Allowed character set/Format
A	Alphanumeric	All alphabetic characters and Spaces
S	Alphanumeric – Special	All alphabetic characters, Space and !@#\$%^&~`()_-+=/\;
N	Numbers	Number 0 to 9
M	Amount	Number 0 to 9 including decimal point where applicable Eg. 150.00
R	Rate	Rate value, Numeric up to specified decimals
D	Date	Date in YYYY-MM-DD format
T	Time	Time in HH:MM:SS format

****NOTE: Above data types will carry maximum length where applicable****

Parameter requirement status

Value	Description
M	Mandatory
O	Optional
C	Conditional

Parameter placement

Value	Description
Q	Query
H	Header
B	Body

Status code category

Value	Description
S	Success
W	Warning
E	Error

HTTP Code Usage

Parameter	Description
200	API call successful. Check the “Status.Code” value for further validation
400	API call failed due to invalid request parameters

1. Internal Transfers

1.1 Funds Transfer [POST]

This service can be used to transfer funds between two accounts. This service optionally supports for a charge.

Request Parameters

Parameter	Req?	Pos.	Format	Description
api_key	M	H		Valid API Key
Account_category	M	B	A 3	Query Using Internal Account(INT) or External Account(EXT)
Source_account_number	M	B	A 13/A 15	Debit account number
Destination_account_number	M	B	A 13/A 15	Credit account number
Transaction_amount	M	B	M	Transaction amount
Debit_transaction_code	O	B	N 3	Debit transaction code
Credit_transaction_code	O	B	N 3	Credit transaction code
User_reference	O	B	A 16	User Reference
Source_account_narration_1	O	B	A 35	Source account narrative
Source_account_narration_2	O	B	A 35	Source account narrative
Source_account_narration_3	O	B	A 35	Source account narrative
Destination_account_narration_1	O	B	A 35	Destination account narrative
Destination_account_narration_2	O	B	A 35	Destination account narrative
Destination_account_narration_3	O	B	A 35	Destination account narrative
Application_type	O	B	A 2	Client application type
Input_branch	O	B	A 4	Posting branch abbreviation
Input_user	O	B	A 4	Posting user
Workstation_id	O	B	A 4	Posting workstation ID
Posting_batch	O	B	A 4	Posting batch ID
Charges_origination_account	O	B	A 3	Charges to originate from account
Charge_code	O	B	A 2	Charge code
Charge_amount	O	B	M	Charge amount

Sample Request

```
https://<<sbk-  
gateway:port>>/Posting/Account/InternalTransfer/1.0/TransferFunds
```

Request payload

```
{  
  "FundsTransfer_Request": {  
    "Account_category": "EXT",  
    "Source_account_number": "003099999999101",  
    "Destination_account_number": "008099999999001",  
    "Transaction_amount": "1000.00",  
    "Debit_transaction_code": "020",  
    "Credit_transaction_code": "520",  
    "User_reference": "TEST001",  
    "Source_account_narration_1": "",  
    "Source_account_narration_2": "",  
    "Source_account_narration_3": "",  
    "Destination_account_narration_1": "",
```

```

    "Destination_account_narration_2": "",
    "Destination_account_narration_3": "",
    "Application_type": "",
    "Input_branch": "MLV ",
    "Input_user": "",
    "Workstation_id": "",
    "Posting_batch": "",
    "Charges_origination_account": "SRC",
    "Charge_code": "EF",
    "Charge_amount": "100"
  }
}

```

Sample Response

```

{
  "FundsTransfer_Response": {
    "Status": {
      "Transaction_Reference": "E200907111040$$AA0000006099547",
      "Timestamp": "2020-09-07T05:40:47.075Z",
      "Code": "0000"
    }
  }
}

```

Status code reference

Status Code	Type	Description
0000	S	Success
1010	E	Internal communication failure with remote host
2020	E	Request rejected by remote host
2030	E	Account validation failed
2070	E	Transaction failed
2090	E	Request contains invalid parameters
2100	E	Account does not exists
2130	E	Restricted Account
2150	E	No sufficient funds
2160	E	Debits not allowed

2. Interbank Transfers

2.1 Initiate CEFTS Transaction [POST]

This service can be used to post charges to customer account using a charge code

Request Parameters

Parameter	Req?	Pos.	Format	Description
api_key	M	H		Valid API Key
Account_category	M	B	A 3	Query Using Internal Account(INT) or External Account(EXT)
Processing_code	M	B	A 6	Processing code
Transaction_code	M	B	A 2	CEFT Transaction code
Source_account_number	M	B	A 13/A 15	Customer source account number
Source_card_number	M	B	A 19	Customer source card number
Source_branch_code	O	B	A 3	Source branch code
Source_customer_name	M	B	A 30	Source customer name
Transaction_amount	M	B	M	Transaction amount
Card_acceptor_terminal_id	O	B	A 8	Card acceptor terminal id
Card_acceptor_id	O	B	A 15	Card acceptor id
Terminal_location	O	B	A 40	Card Terminal location
Destination_account_number	M	B	A 20	Destination account number
Destination_card_number	M	B	A 19	Destination card number
Destination_bank_code	M	B	A 4	Destination bank code
Destination_customer_name	M	B	A 30	Destination customer name
Destination_branch_code	O	B	A 3	Destination branch code
Currency_code	M	B	A 3	Currency code
Reference	M	B	A 20	Transaction reference
Particular_details	O	B	A 20	Particulars
Additional_data	O	B	A 200	Additional data
Authorized_user	O	B	A 4	Transaction authorization user
Input_branch	O	B	A 4	Input branch
Input_user	O	B	A 4	Transaction input user
Workstation_id	O	B	A 4	Input workstation id
Posting_batch	O	B	A 4	Input posting batch
Charge_code	O	B	A 2	Charge code
Charge_amount	O	B	M	Charge amount
Charges_credit_branch	O	B	A 4	Charges credit branch
Charges_narrative_1	O	B	A 35	Charges narrative 1
Charges_narrative_2	O	B	A 35	Charges narrative 2
Charges_narrative_3	O	B	A 35	Charges narrative 3

Processing_code

Processing Code is a series of digits that describe the effect of a transaction on the customer account and the type of accounts affected.

- Positions 1-2 Transaction Type: a two-digit code identifying the transaction type.
- Positions 3-4, Account Type (from): a two-digit code identifying the account type affected by this transaction.

- Positions 5-6, Account Type (to): a two-digit code identifying the account type affected by this transaction.

Transaction Type
48: EFT Credit Transaction
49: EFT Debit Transaction
40: EFT Off-Us Transaction
41/21: EFT Debit-Credit Transaction
42: EFT Debit Transaction (initiated from any channel by Issuer)

Account Type (from)	Account Type (to)
00: not specified	00: not specified
10: savings account	10: savings account
20: checking account	20: checking account
30: credit card account	30: credit card account

Sample Request

```
https://<<sbk-  
gateway:port>>/Posting/Account/Cefts/1.0/InitiateCEFTSTransfer
```

Request Payload

```
{
  "CEFTSTransactionRequest": {
    "Processing_code": "482000",
    "Transaction_code": "52",
    "Transaction_amount": "100",
    "Card_acceptor_terminal_id": "",
    "Card_acceptor_id": "",
    "Terminal_location": "",
    "Channel_type": "ANY",
    "Account_category": "EXT",
    "Source_account_number": "030100088888004",
    "Source_card_number": "",
    "Source_customer_name": "ABC Collections",
    "Source_bank_code": "6287",
    "Source_branch_code": "",
    "Source_wallet_number": "",
    "Destination_card_number": "",
    "Destination_account_number": "123456789012",
    "Destination_bank_code": "6990",
    "Destination_customer_name": "TEST",
    "Destination_branch_code": "",
    "Destination_wallet_number": "",
    "Currency_code": "LKR",
    "Reference": "Test_Ref_0001",
    "Particular_details": "",
    "Additional_data": "",
    "Authorized_user": "",
    "Customer_account_narration_1": "",
    "Customer_account_narration_2": "",
    "Customer_account_narration_3": "",
    "Internal_account_narration_1": ""
  }
}
```

```

        "Internal_account_narration_2": "",
        "Internal_account_narration_3": "",
        "Application_type": [],
        "Input_branch": [],
        "Input_user": [],
        "Workstation_id": [],
        "Posting_batch": [],
        "Charges_origination_account": [],
        "Charge_code": [],
        "Charge_amount": [],
        "Charges_credit_branch": [],
        "Charges_narrative_1": [],
        "Charges_narrative_2": [],
        "Charges_narrative_3": [],
        "Existing_hold_amount": [],
        "Hold_reference": []
    }
}

```

Sample Response

```

{
  "CEFTSTransactionResponse": {
    "Status": {
      "Transaction_Reference": "E210209155449$$AA0000009826783",
      "Timestamp": "2021-02-09T10:24:51.865Z",
      "Code": "0000",
      "Message": "",
      "Description": ""
    },
    "CEFTSTransaction_Detail": {
      "Transaction_Date": "2021-02-02",
      "Transaction_id": "013210821730",
      "Approval_number": "145512",
      "System_trace_audit_number": "5726222275",
      "Response_code": "00",
      "Response_code_desc": "Transaction Successful"
    }
  }
}

```

Status code reference

Status Code	Type	Description
0000	S	Success
1010	E	Internal communication failure with remote host
2020	E	Request rejected by remote host
2030	E	Account validation failed
2070	E	Transaction failed
2090	E	Request contains invalid parameters
2100	E	Account does not exists
2130	E	Restricted Account
2150	E	No sufficient funds

3. Account Inquiry

3.1 Account Balance Inquiry [GET]

This service returns balance and related information for an account.

Request Parameters

Parameter	Req?	Type	Format	Description
api_key	M	H		Valid API Key
AccountCategory	M	Q	A 3	Query Using Internal Account(INT) or External Account(EXT)
AccountNumber	M	Q	A 13/A 15	Account Number in respective account category

Sample Request

```
https://<<sbk-gateway:port>>/Inquiry/Account/AccountInquiry/1.0/GetAccountBalance?AccountCategory=EXT&AccountNumber=001001338903101
```

Sample Response

```
{
  "Account_Balance_Inquiry": {
    "Status": {
      "Transaction_Reference": "E27123184325$$AA00XXX3096396",
      "Timestamp": "2020-08-23T13:13:27.006Z",
      "Code": "0000",
      "Message": null,
      "Description": null
    },
    "Account": {
      "Account_branch": "0010",
      "Basic_part_of_account_number": "X99003",
      "Account_suffix": "101",
      "Account_short_name": "PERERA A",
      "Arabic_short_name": "PERERA A",
      "Customer_mnemonic": "X99003",
      "Customer_location": "SIA",
      "Customer_full_name": "ANIL PERERA",
      "Arabic_customer_full_name": "ANIL PERERA",
      "Account_Type": "EA",
      "Owning_application_code": null,
      "Account_structure": "M",
      "Currency_mnemonic": "LKR",
      "Currency_edit_field": "2",
      "Internal_non_customer_account": "N",
      "Deceased_or_liquidated": "N",
      "Account_status_Blocked": "N",
      "Account_status_Inactive": "N",
      "Account_closing": "N",
      "Overdraft_limit": "0",
      "Limit_expiry_date": null,
      "Overdraft_expired": "N",
      "Reserved_balance": "4700",
      "Ledger_balance": "143697.99",
      "Status_balance": "143697.99",
    }
  }
}
```

```

    "Previous_cleared_balance": "143697.99",
    "Current_cleared_balance": "143697.99",
    "Current_available_balance": "138997.99",
    "Is_Current_available_balance": "Y",
    "Later_available_balance": "138997.99",
    "Is_Later_available_balance": "Y",
    "Next_available_balance_1": "138997.99",
    "Next_available_balance_2": "0",
    "Is_Next_available_balance": "Y",
    "Forward_balance": "138997.99",
    "Is_Forward_balance": "Y",
    "Clearing_today": "0",
    "Clearing_later_today": "0",
    "Clearing_tomorrow": "0",
    "Clearing_after_tomorrow": "0",
    "Shadow_debits": "0",
    "Shadow_credits": "0",
    "Shadow_adjustments": "0",
    "OD_against_uc_cheques": "N",
    "Susp_OD_UC_cheques": "N",
    "Notice_AC_special_condition": "N",
    "Withdrawals_maturing_today": "NaN",
    "Closing_withdrawal": null,
    "Withdrawals_maturing_tomorrow": "NaN",
    "Closing_withdrawal_1": null,
    "Withdrawals_maturing_after_tomorrow": "NaN",
    "Closing_withdrawal_2": null,
    "Hold_queued_amount": "0",
    "Customer_Group": null,
    "Real_time_balance_order": null,
    "Ext_Account_No": "001001338903101",
    "Edited_Ext_Account_No": "0010-01338903-101",
    "International_bank_account": null,
    "Edited_international_bank": null,
    "Date_account_opened": "2011-07-05",
    "Date_account_closed": null,
    "Customer_type": "EZ",
    "SCL_Credit_authorized_amount": "0",
    "SCL_Credit_not_authorized_amount": "0",
    "SCL_Credit_returned_amount": "0",
    "SCL_Debit_authorized_amount": "0",
    "SCL_Debit_not_authorized_amount": "0",
    "OCL_Unrealized_cheques_amount": "0",
    "OCL_Clearing_today_amount": "0",
    "OCL_Purchased_cheques_amount": "0",
    "OCL_Returned_cheques_amount": "0"
  }
}
}

```

Status code reference

Status Code	Type	Description
0000	S	Success
1010	E	Internal communication failure with remote host
2010	E	Record not available. Invalid search key.
2030	E	Account validation failed

3.2 Account Transaction History Inquiry [GET]

This service returns transaction history for an account and can be invoked by either of below methods.

1. Get n number of recent transactions
To get the last 'n' transactions, [NumberOfTransactions] parameter must be populated and [StartDate], [EndDate] is not required
2. Get transactions for a particular time period
Provide the [StartDate] and [EndDate] parameters without [NumberOfTransactions]

When using the API with option 2, there can be instances where number of transaction are too large to return via single call. In such scenario, API response' [Date_to] field will contain the last transaction date from the response. To obtain the next transaction set, call the API again using start date [Date_to] + 1

Request Parameters

Parameter	Req?	Type	Format	Description
api_key	M	H		Valid API Key
AccountCategory	M	Q	A 3	Query Using Internal Account(INT) or External Account(EXT)
AccountNumber	M	Q	A 13/A 15	Account Number in respective account category
StartDate	C	Q	D	Start date of Transaction history
EndDate	C	Q	D	End date of Transaction history
NumberOfTransactions	C	Q	N 5	Number of recent transactions to retrieve

Sample Request

```
https://<<sbk-gateway:port>>/Inquiry/Account/AccountInquiry/1.0/GetAccountTransactions?AccountCategory=EXT&AccountNumber=008000999999001&StartDate=2021-01-01&EndDate=2021-01-01
```

Sample Response

```
{
  "TransactionHistoryInquiryResponse": {
    "Status": {
      "Transaction_Reference": "E210309125555$$AA0000005403865",
      "Timestamp": "2021-03-09T07:25:55.902Z",
      "Code": "0000",
      "Message": null,
      "Description": null
    },
    "AccountSummary": {
      "Account_branch": "0080",
      "Account_basic_number": "102547",
      "Account_suffix": "001",
      "Account_short_name": "ABES PLC",
      "Customer_mnemonic": "102547",
      "Customer_location": "ABA",
      "Account_type": "CA",
      "Owning_application_code": {},
      "Account_structure": "M",

```



```

    "Currency_mnemonic": "LKR",
    "Currency_decimals": "2",
    "Internal_non_customer_account": "N",
    "Deceased_or_liquidated": "N",
    "Account_blocked": "N",
    "Account_inactive": "N",
    "Account_closed": "N",
    "Contingent_account": "N",
    "Date_from": "2020-12-09",
    "Date_to": "2021-03-09",
    "Starting_balance": "4904460.16",
    "Number_of_recent_transactions": "0",
    "Statement_number": "0",
    "Available_balance": "4903760.16"
  },
  "Transaction": [
    {
      "Posting_date": "2021-02-16",
      "Value_date": "2021-02-16",
      "Posting_currency": "LKR",
      "Posting_currency_decimals": "2",
      "Transaction_code": "020",
      "Transaction_code_mnemonic": "TRF",
      "Transaction_Code_Name": "TRANSFER - DEBIT",
      "Input_branch_mnemonic": "MLV",
      "Account_branch_mnemonic": "KPY",
      "Posting_group": "$$AA",
      "Posting_amount": "-100.00",
      "Print_transaction_code_name_as_narrative": "N",
      "Transaction_Status": "D",
      "Users_own_reference": "TEST001",
      "Sundry_reference_code": null,
      "User_code_1": null,
      "User_code_2": null,
      "Number_of_items_within_this_posting": "001",
      "Statement_number": "0000323",
      "Narrative_record_exists": "Y",
      "Narrative_1": null,
      "Narrative_2": null,
      "Narrative_3": null,
      "Narrative_4": "E210216225125$$AA0000006409366",
      "Additional_line_1": null,
      "Additional_line_2": null,
      "Additional_line_3": null,
      "Additional_line_4": null,
      "Additional_line_5": null,
      "Additional_line_6": null,
      "Original_currency": null,
      "Original_currency_decimals": "2",
      "Original_amount": "0",
      "Posting_sequence_number": "0000015",
      "Running_balance": "4904360.16",
      "Event_key": "12102168276330000001",
      "Available_Date": null,
      "Timestamp": "2021-02-16-22.50.41.782000"
    },
    {
      "Posting_date": "2021-02-16",
      "Value_date": "2021-02-16",
      "Posting_currency": "LKR",
      "Posting_currency_decimals": "2",

```

```

    "Transaction_code": "020",
    "Transaction_code_mnemonic": "TRF",
    "Transaction_Code_Name": "TRANSFER - DEBIT",
    "Input_branch_mnemonic": "MLV",
    "Account_branch_mnemonic": "KPY",
    "Posting_group": "$$AA",
    "Posting_amount": "-100.00",
    "Print_transaction_code_name_as_narrative": "N",
    "Transaction_Status": "D",
    "Users_own_reference": "TEST001",
    "Sundry_reference_code": null,
    "User_code_1": null,
    "User_code_2": null,
    "Number_of_items_within_this_posting": "001",
    "Statement_number": "0000323",
    "Narrative_record_exists": "Y",
    "Narrative_1": null,
    "Narrative_2": null,
    "Narrative_3": null,
    "Narrative_4": "E210216225225$$AA0000009419309",
    "Additional_line_1": null,
    "Additional_line_2": null,
    "Additional_line_3": null,
    "Additional_line_4": null,
    "Additional_line_5": null,
    "Additional_line_6": null,
    "Original_currency": null,
    "Original_currency_decimals": "2",
    "Original_amount": "0",
    "Posting_sequence_number": "0000017",
    "Running_balance": "4904260.16",
    "Event_key": "12102168276380000001",
    "Available_Date": null,
    "Timestamp": "2021-02-16-22.51.41.825000"
  }
}
}
}

```

Status code reference

Status Code	Type	Description
0000	S	Success
1010	E	Internal communication failure with remote host
2010	E	Record not available. Invalid search key.
2030	E	Account validation failed
2060	E	Request Exceeds the maximum range allowed

4. JustPay Acquirer Services

4.1 JustPay Account Registration [POST]

This service onboard account (both SBK and Non SBK) for justpay transactions.

Request Parameters

Parameter	Req?	Type	Format	Description
api_key	M	H		Valid API Key
Justpay_code	M	B	A 30	Justpay code for SBK/APP
Device_id	M	B	A 8 - A 45	Device Id
User_id	M	B	A 20	Accepts in following format N:922201013V N:922201013X N:19922201013,P:MD3942834 P:N3942834
NIC_number	M	B	A 12	NIC number of the customer
User_name	M	B	A 3 – A 50	User Name: Accepts only letters, full stop and space
Platform	M	B	A 15	Application platform. Accepts following format. android, ios
Mobile_number	M	B	A 15	Mobile number in following format 0094XXXXXXXXXX
Email_address	M	B	A 50	Email address
Retrieval_reference	O	B	A 30	Request reference. If not provided, Reference generated from the API will be assigned.
Originating_branch_code	O	B	A 4	Originating branch code
Destination_bank_code	M	B	A 4	Destination bank code
Destination_branch_code	O	B	A 4	Destination branch code
Account_number	M	B	A 20	Account number
Account_title	M	B	A 40	Account title
Currency	M	B	A 3	Currency
Input_user	O	B	A 4	Input User
Input_branch	O	B	A 4	Input branch abbreviations
Application_type	O	B	A 4	Application type code
Workstation_id	O	B	A 4	Workstation ID

Sample Request

```
https://<<sbk-  
gateway:port>>/JustPay/Acquirer/1.0/JustPayRegisterAccount
```

Request Payload

```
{  
  "JustPayRegisterAccount_Request": {  
    "Justpay_code": "6289_M001_001",  
    "Device_id": "abcde123-ab12-a123-ab12-abcdef123456",  
    "User_id": "N:833333999V",  
    "NIC_number": "833333999V",  
    "User_name": "ABC PE",  
    "Platform": "ANDROID",
```

```

    "Mobile_number": "94777339999",
    "Email_address": "mail@google.lk",
    "Retrieval_reference": "",
    "Originating_branch_code": "050",
    "Destination_bank_code": "6990",
    "Destination_branch_code": "001",
    "Account_number": "123456789",
    "Account_title": "TEST",
    "Currency": "LKR",
    "Application_type": "",
    "Input_branch": "",
    "Input_user": "",
    "Workstation_id": "",
    "Posting_batch": ""
  }
}

```

Sample Response

```

{
  "JustPayRegisterAccount_Response": {
    "Status": {
      "Transaction_Reference": "E210506024748$$AA0000004141113",
      "Timestamp": "2021-05-05T21:18:04.989Z",
      "Code": "0000",
      "Message": null,
      "Description": null
    }
  }
}

```

Status code reference

Status Code	Type	Description
0000	S	Success
1010	E	Internal communication failure with remote host
2030	E	Account validation failed
2100	E	Account does not exists
2110	E	Customer Information not valid
2120	E	Product not eligible
2130	E	Restricted account
2140	E	Customer not eligible
3020	E	Issuer Unavailable
3030	E	Transaction declined by Issuer

4.2 Verify JustPay Registration [GET]

This service validates the JustPay account registration against the OTP entered by the App User. If the tokenization is enabled for an application, this service will optionally return a token number for the registered account. In such scenario transactions will not be allowed using clear numbers

Request Parameters

Parameter	Req?	Type	Format	Description
api_key	M	H		Valid API Key
RequestId	M	Q	A 30	Retrieval reference from registration request or Transaction reference generated by API (if retrieval reference not available)
RegistrationOTP	M	Q	A 4	JustPay registration OTP

Sample Request

```
https://<<sbk-gateway:port>>/JustPay/Acquirer/1.0/VerifyJustPayRegistration?RequestId=E210506024748$$AA0000004141113&RegistrationOTP=0531
```

Sample Response

```
{
  "JustPayAccountValidation_Response": {
    "Status": {
      "Transaction_Reference": "E210629182927$$AA0000006055217",
      "Timestamp": "2021-06-29T12:59:28.034Z",
      "Code": "0000",
      "Message": null,
      "Description": null
    },
    "Account_token": "@TER1TBEVST0XXX8BO1E",
    "Account_mask": "92*****193"
  }
}
```

Status code reference

Status Code	Type	Description
0000	S	Success
1010	E	Internal communication failure with remote host
2020	E	Request rejected by remote host. - Remote server returned failed status. Refer Description field for more information
2200	E	Registration request expired
2210	E	Transaction OTP is incorrect

4.3 Create JustPay Certificate [POST]

This service is used to request a JustPay certificate for a new user registration.

Request Parameters

Parameter	Req?	Type	Format	Description
api_key	M	H		Valid API Key
Justpay_code	M	B	A 30	Justpay code provided by SBK
User_id	M	B	A 20	User Id
User_name	M	B	A 50	Customer registered user name
Mobile_no	M	B	A 15	Customer mobile number Ex. 0094772233444
Email_address	M	B	A 50	Email address
Platform	M	B	A 15	Platform as in registration request Ex. "Web"
Retrieval_reference	M	B	A 30	Transaction_Reference value received for JustPay registration
Application_type	O	B	A 4	Application type
Input_branch	O	B	A 4	Input branch
Input_user	O	B	A 4	Input user
Workstation_id	O	B	A 4	Workstation id
Posting_batch	O	B	A 4	Posting batch

Sample Request

```
https://<<sbk-  
gateway:port>>/JustPay/Acquirer/1.0/JustPayGetCertificate
```

Sample Payload

```
{  
  "JPCertificateIssue_request": {  
    "Justpay_code": "6289_M91495_40056",  
    "User_id": "N:832854141V",  
    "User_name": "ABC PE",  
    "Mobile_no": "0094777339819",  
    "Email_address": "abc_pe@gmail.lk",  
    "Platform": "WEB",  
    "Retrieval_reference": "E230731144431$$AA0000006518639",  
    "Application_type": "",  
    "Input_branch": "",  
    "Input_user": "",  
    "Workstation id": "",  
    "Posting_batch": ""  
  }  
}
```

Sample Response

```
{  
  "JPCertificateIssue_response": {  
    "Status": {  
      "Transaction_Reference": "E210506175209$$AA0000009027209",  
      "Timestamp": "2021-05-06T12:22:09.922Z",  
      "Code": "0000",  
    }  
  }  
}
```

```

    "Message": null,
    "Description": null
  }
}
}

```

Status code reference

Status Code	Type	Description
0000	S	Success
1010	E	Internal communication failure with remote host
2020	E	Request rejected by remote host. - Remote server returned failed status. Refer Description field for more information

4.4 JustPay Sign Digital Mandate [POST]

This service is used to sign the user mandate with the certificate created at LPPL.

Request Parameters

Parameter	Req?	Type	Format	Description
api_key	M	H		Valid API Key
Justpay_code	M	B	A 30	Justpay code for SBK/APP
User_id	M	B	A 20	User id as in registration request
User_name	M	B	A 50	Customer registered user name
Mobile_no	M	B	A 15	Mobile number in following format 0094XXXXXXXXXX
Email_address	M	B	A 50	Email address
Platform	M	B	A 15	Platform
Retrieval_reference	M	B	A 30	Transaction reference received from registration request
Message_to_sign	M	B	A	Message to be signed
Application_type	O	B	A 4	Application type
Input_branch	O	B	A 4	Input branch
Input_user	O	B	A 4	Input user
Workstation_id	O	B	A 4	Workstation ID
Posting_batch	O	B	A 4	Posting batch

Sample Request

```

https://<<sbk-
gateway:port>>/JustPay/Acquirer/1.0/JustPaySignDigitalMandate

```

Request Payload

```

{
  "JPSignMessage_request": {
    "Justpay_code": "6289_M91495_056",

```

```

    "User_id": "N:833339999V",
    "User_name": "ABC_PE",
    "Mobile_no": "0094774455666",
    "Email_address": "abc_pe@gmail.lk",
    "Platform": "WEB",
    "Retrieval_reference": "E230731144431$$AA0000006518639",
    "Message_to_sign": "<< Message to be signed >>",
    "Application_type": "",
    "Input_branch": "",
    "Input_user": "",
    "Workstation_id": "",
    "Posting_batch": ""
  }
}

```

Sample Response

```

{
  "JPSTransaction_response": {
    "Status": {
      "Transaction_Reference": "E230731154724$$AA0000009968362",
      "Timestamp": "2023-07-31T10:17:36.233Z",
      "Code": "0000",
      "Message": null,
      "Description": null
    },
    "Signed_message":
    "MIAGCSqGSib3DQEHAQCAMIACAQExDTALBglghkgBZQMEAgMwgAYJKoZIhvcNAQcBoIAEP
    lRlc3QgTWVzc2FnZSB0byBiZSBzaWduZWQuIEFncmjsOYUn+PCszMloIRQOKvoTl9uXeeV
    P2eTUgspUcRE5+HQpbRaNTktsxP03+CT6GyQ9Y0Hf6AaWsRgpUmh30IkhRW+aU3bQx9Zje
    WLpiW69slr6tn8ZFXg6Qfsw2FIRahfyF60TVzJUz4ibYc/FEPka4AAAAA="
  }
}

```

Status code reference

Status Code	Type	Description
0000	S	Success
1010	E	Internal communication failure with remote host
2020	E	Account validation failed
7200	E	Certificate has been expired & renewal failed

4.5 Initiate JustPay Transaction [POST]

This service initiates a JustPay Transaction

Request Parameters

Parameter	Req?	Type	Format	Description
api_key	M	H		Valid API Key
Justpay_code	M	B	A 30	Justpay code for SBK/APP

Device_id	M	B	A 8 - A 45	Device Id as in registration request
User_id	M	B	A 20	User id as in registration request
Retrieval_reference	O	B	A 30	Request reference. If not provided, Reference generated from the API will be assigned.
NIC_number	M	B	A 12	NIC number as in registration request
Originating_branch_code	O	B	A 4	Transaction origination branch code
Originating_account_number	M	B	A 15	Transaction origination account number
Originating_account_name	M	B	A 35	Transaction origination account name
Destination_bank_code	M	B	A 4	Destination bank code
Destination_branch_code	O	B	A 4	Destination branch code
Destination_account_number	M	B	A 15	Destination account number token returned at the time of verifying the Justpay Registration (API 4.2)
Destination_account_name	M	B	A 35	Destination account name
Transaction_narration	O	B	A 50	Transaction narration
Transaction_amount	M	B	M	Transaction amount
Settlement_act_narration_1	O	B	A 35	Settlement account narration 1
Settlement_act_narration_2	O	B	A 35	Settlement account narration 2
Settlement_act_narration_3	O	B	A 35	Settlement account narration 3
Parking_act_narration_1	O	B	A 35	Parking account narration 1
Parking_act_narration_2	O	B	A 35	Parking account narration 2
Parking_act_narration_3	O	B	A 35	Parking account narration 3
Currency	M	B	A 3	Currency
Input_user	O	B	A 4	Input User
Input_branch	O	B	A 4	Input branch abbreviations
Application_type	O	B	A 4	Application type code
Workstation_id	O	B	A 4	Workstation ID
Posting_batch	O	B	A 4	Posting Batch

Sample Request

```
https://<<sbk-  
gateway:port>>/JustPay/Acquirer/1.0/InitiateJustPayTransaction
```

Request Payload

```
{
  "JustPayTransaction_Request": {
    "Justpay_code": "6289_M001_001",
    "Device_id": "abcde123-ab12-a123-ab12-abcdef123456",
    "User_id": "N:833339999V",
    "Retrieval_reference": "",
    "NIC_number": "833339999V",
    "Originating_branch_code": "050",
    "Originating_account_number": "003009999999100",
    "Originating_account_name": "Test1",
    "Destination_bank_code": "6990",
    "Destination_branch_code": "001",
    "Destination_account_number": "@TER1TBEVST0XXX8B01E",
    "Destination_account_name": "Test D",
    "Transaction_narration": "Transaction_narration",
    "Transaction_amount": "500.00",
```

```

    "Settlement_act_narration_1": "",
    "Settlement_act_narration_2": "",
    "Settlement_act_narration_3": "",
    "Settlement_act_narration_4": "",
    "Parking_act_narration_1": "",
    "Parking_act_narration_2": "",
    "Parking_act_narration_3": "",
    "Parking_act_narration_4": "",
    "Currency": "LKR",
    "Application_type": "",
    "Input_branch": "",
    "Input_user": "",
    "Workstation_id": "",
    "Posting_batch": ""
  }
}

```

Sample Response

```

{
  "JustPayTransaction_Response": {
    "Status": {
      "Transaction_Reference": "E210506180954$$AA0000008654715",
      "Timestamp": "2021-05-06T12:39:59.537Z",
      "Code": "0000",
      "Message": null,
      "Description": null
    },
    "RRN": "0000000460123",
    "STAN": "000111",
    "Parking_account_ext": null,
    "Parking_account_int": null
  }
}

```

Status code reference

Status Code	Type	Description
0000	S	Success
0510	E	Transaction failed - Refund action failed - Followup required
1010	E	Internal communication failure with remote host
2020	E	Account validation failed
3010	E	Transaction failed - Refund action successful
3020	E	Issuer unavailable
3030	E	Transaction declined by issuer

4.6 Get JustPay Transaction Status [GET]

This service returns the status of previously processed transaction and this must be used to check the status if the transaction service did not respond to the request.

Request Parameters

Parameter	Req?	Type	Format	Description
api_key	M	H		Valid API Key
JustpayCode	M	Q	A 30	Justpay code for APP

RetrievalReference	M	Q	A 30	Retrieval reference from registration request or Transaction reference generated by API (if retrieval reference not provided)
--------------------	---	---	------	---

Sample Request

```
https://<<sbk-gateway:port>>/JustPay/Acquirer/1.0/GetJustPayTransactionStatus?JustpayCode=6289_M001_001&RetrievalReference=E210506180954$$AA0000008654715
```

Sample Response

```
{
  "JustPayTransactionStatus_Response": {
    "Status": {
      "Transaction_Reference": "E210506181222$$AA0000006926794",
      "Timestamp": "2021-05-06T12:42:23.788Z",
      "Code": "2020",
      "Message": "Request rejected by remote host",
      "Description": "09:Request in progress No Original transaction"
    },
    "Original_transaction_reference": "E220127145854$$AA0000008725939",
    "RRN": "000000460123",
    "STAN": "000111",
    "Transaction_date": "2022-01-27"
  }
}
```

Status code reference

Status Code	Type	Description
0000	S	Success
1010	E	Internal communication failure with remote host
2020	E	Request rejected by remote host. - Remote server returned failed status. Refer Description field for more information

4.7 Refund JustPay Transaction [POST]

This service initiates a JustPay refund Transaction

Request Parameters

Parameter	Req?	Type	Format	Description
api_key	M	H		Valid API Key
Justpay_code	M	B	A 30	Justpay code for SBK/APP
Device_id	M	B	A 8 - A 45	Device Id as in registration request
User_id	M	B	A 20	User id as in registration request
Retrieval_reference	M	B	A 30	Request reference.
Input_user	O	B	A 4	Input User

Input_branch	O	B	A 4	Input branch abbreviations
Application_type	O	B	A 4	Application type code
Workstation_id	O	B	A 4	Workstation ID
Posting_batch	O	B	A 4	Posting Batch

Sample Request

```
https://<<sbk-  
gateway:port>>/JustPay/Acquirer/1.0/RefundJustPayTransaction
```

Request Payload

```
{  
  "JustPayRefundTransaction Request": {  
    "Justpay_code": "6289_M001_001",  
    "Device_id": "abcde123-ab12-a123-ab12-abcdef123456",  
    "User_id": "N:833339999V",  
    "Retrieval_reference": "E210506180954$$AA0000008654715",  
    "Application_type": "",  
    "Input_branch": "",  
    "Input_user": "",  
    "Workstation_id": "",  
    "Posting_batch": ""  
  }  
}
```

Sample Response

```
{  
  "JustPayRefundTransaction_Response": {  
    "Status": {  
      "Transaction_Reference": "E210512100708$$AA0000008259649",  
      "Timestamp": "2021-05-12T04:37:15.063Z",  
      "Code": "0000",  
      "Message": null,  
      "Description": null  
    }  
  }  
}
```

Status code reference

Status Code	Type	Description
0000	S	Success
1010	E	Internal communication failure with remote host
2020	E	Account validation failed

5. LankaQR Acquirer Services

5.1 Initiate LankaQR Transaction [POST]

This service will initiate a LankaQR transaction using JustPay registered account

Request Parameters

Parameter	Req?	Type	Format	Description
api_key	M	H		Valid API Key
Justpay_code	M	B	A 30	Justpay code for SBK/APP
Device_id	M	B	A 8 - A 45	Device Id
Justpay_user_id	M	B	A 20	Accepts in following format N:922201013V N:922201013X N:19922201013,P:MD3942834 P:N3942834
Institution_id	M	B	N 3	User the fixed value "1"
Party_id	M	B	N 15	Unique id assigned to channel application. Channel will receive this credential from Bank or over auto-generated email
Function	M	B	A 30	Use the fixed value "ThirdPartyScanNPay"
Channel_user_id	M	B	A 100	Channel user Id
Channel_password	M	B	A 100	Channel Password
Payment_mode	O	B	A 30	Use the fixed value "A"
Retrieval_reference	M	B	A 12	Unique request reference number
Transation_sequence	M	B	N 12	Transaction unique sequence
QR_payload	M	B	A 512	QR Payload
Customer_bank_code	M	B	A 4	Customer bank code
Customer_account	M	B	A 20	Customer account number
Customer_name	M	B	A 50	Customer name
Merchant_pan	M	B	A 28	Merchant Pan TAG 26 value need to be passed
Merchant_bank_code	M	B	A 4	Merchant bank code
Merchant_name	M	B	A 25	Merchant name
Transaction_ccy	M	B	A 3	Transaction currency code
Transaction_ccy_num	M	B	N 3	Transaction currency number
Transaction_amount	M	B	M 12	Transaction amount
Tip_indicator	O	B		Tip Indicator 01 – Tip Amount need to be passed 02 – Convenience fee 03 – Convenience fee percentage
Tip_amount	O	B	M 12	Tip Amount in case of Tip_indicator ='01'
Convenience_fee	O	B	M 12	Convenience fee amount or percentage in case of Tip_indicator is '02', '03'
Merchant_catg_code	M	B	A 4	Merchant category code
Merchant_city	M	B	A 25	Merchant city
Country_code	M	B	A 5	Merchant country code
Postal_code	M	B	N 6	Postal code
Transaction_desc	M	B	A 50	Transaction Description/Narration

Payment_ref1_code	M	B		If Tag62 is present in QR then this should be sent Possible values 01 - BILL_NUMBER 02 - MOBILE_NUMBER 03 - STORE_ID 04 - LOYALTY_NUMBER 05 - REFERENCE_ID 06 - CONSUMER_ID 07 - TERMINAL_ID 08 – PURPOSE
Payment_ref1	M	B		If Tag62 is present in QR then this should be sent
Payment_ref2_code	O	B		If Tag62 is present in QR then this should be sent Possible values 01 - BILL_NUMBER 02 - MOBILE_NUMBER 03 - STORE_ID 04 - LOYALTY_NUMBER 05 - REFERENCE_ID 06 - CONSUMER_ID 07 - TERMINAL_ID 08 – PURPOSE
Payment_ref2	O	B		If Tag62 is present in QR then this should be sent
Transaction_date	M	B	D	Transaction date
Transaction_time	M	B	T	Transaction time
IP_address	O	B	A 50	IP Address
Session_id	O	B	A 50	Session Id
Check_value	M	B	A 500	Check value for the request. Calculation method is included below

Check value calculation

Note:
Data string used for the check value calculation should be constructed as below
Channel_user_id|Retrieval_reference|Function| |Payment_mode|Transaction_amount|Merchant_name

Calculation method

```
import javax.crypto.Mac;
import javax.crypto.SecretKey;
import javax.crypto.spec.SecretKeySpec;
public class GenerateChecksum
{
    private final char[] HEX_TABLE = { '0', '1', '2', '3', '4', '5',
    '6', '7', '8', '9', 'A', 'B', 'C', 'D', 'E', 'F' };
    //secret Key - Key shared between channel and Euronet
    //Data String is pipe delimited values
```

```

    public String getSecureSHAHash(String secretKey, String dataString)
    throws Exception
    {
        byte[] mac = null;
        SecretKey key = new SecretKeySpec(secretKey.getBytes("UTF-
8"), "HmacSHA512");
        Mac m = Mac.getInstance("HmacSHA512");
        m.init(key);
        m.update(dataString.getBytes("UTF-8"));
        mac = m.doFinal();
        String hashValue = hex(mac);
        return hashValue;
    } //End of hex

    public String hex(byte[] input) throws Exception
    {
        StringBuffer sb = new StringBuffer(input.length * 2);
        for (int i = 0; i < input.length; i++)
        {
            sb.append(this.HEX_TABLE[(input[i] >> 4 & 0xF)]);
            sb.append(this.HEX_TABLE[(input[i] & 0xF)]);
        }
        return sb.toString();
    }
}

```

Finally, the calculated checksum should be encoded with Base64

Sample Request

<https://dev.apigateway1.seylan.lk:2250/QR/LankaQR/1.0/InitiateLankaQRTransaction>

Request Payload

```

{
  "LankaQRTransaction_Request": {
    "Justpay_code": "6289_M001_001",
    "Justpay_user_id": "N:8399999991V",
    "Device_id": "ABCDE123-AB12-A123-AB12-ABCDEF123456",
    "Institution_id": "1",
    "Party_id": "920590000000001",
    "Function": "ThirdPartyScanNPay",
    "Channel_user_id": "API_Indika88732",
    "Channel_password": "t#N-f890fs0s",
    "Payment_mode": "A",
    "Retrieval_reference": "202202210089",
    "Transaction_sequence": "202202210089",
    "QR_payload":
    "00020155020101021102164157241220007614031641572412200076145802LK04155
419022220007612632002816288000162880000622000761705921LANKA      FILLING
STATION0515541902222000761801000060000006010COLOMBO
0261050020062070503***52045542530314464450002EN0121LANKA      FILLING
STATION0210COLOMBO 02630420CF",
    "Customer_bank_code": "6990",
    "Customer_branch_code": "",
    "Customer_account": "@TZ3CL05V4RRTHPLZTK7",
    "Customer_name": "TheCustomer",
    "Merchant_pan": "16990007000000000000044680000",
    "Merchant_bank_code": "6288",
    "Merchant_name": "The Shop",
    "Transaction_ccy": "LKR",
    "Transaction_amount": "150",
  }
}

```

```

    "Tip_indicator": "01",
    "Tip_amount": "10",
    "Convenience_fee": "",
    "Merchant_catg_code": "4201",
    "Merchant_city": "COLOMBO 02",
    "Country_code": "LK",
    "Postal_code": "",
    "Payment_refl_code": "03",
    "Payment_refl": "Test",
    "Payment_ref2_code": "05",
    "Payment_ref2": "07183332989",
    "Transaction_date": "2022-03-11",
    "Transaction_time": "13:51:12",
    "Transaction_desc": "Test QR",
    "IP_address": "",
    "Session_id": "",
    "Check_value":
"RTI3MUE3MTYyNkU1QkUwNTI0Q0M5RDBGQThCODNFRTdCOUFFQ0ZEMTRFMzhEM0Y4RTk1R
jg0NTRDOEIxODI3NUQ5M0Q3Qjk0RjkzMz1BQkEyNDI4NUM0MjE0MENENkYxNDg2MTA0Q0V
FNUEwRjc4RTg0MTU1Mz1FODg0MkVFNzY=",
  }
}

```

Sample Response

```

{
  "LankaQRTransaction_Response": {
    "Status": {
      "Transaction_Reference": "E220310121313$$AA0000009611121",
      "Timestamp": "2022-03-10T06:43:15.647Z",
      "Code": "0000",
      "Message": null,
      "Description": null
    },
    "Debit_RRN": null,
    "Debit_STAN": null,
    "Credit_RRN": null,
    "Credit_STAN": null
  }
}

```

Status code reference

Status Code	Type	Description
0000	S	Success
1010	E	Internal communication failure with remote host
2020	E	Request rejected by remote host
2070	E	Transaction failed
2100	E	Account does not exists.
2150	E	No sufficient funds
2130	E	Restricted Account
3020	E	Issuer unavailable
3030	E	Request declined by issuer

5.2 Inquire LankaQR Transaction Status [GET]

This service returns the status of previously processed transaction and this must be used to check the status if the transaction service did not respond to the request.

Request Parameters

Parameter	Req?	Type	Format	Description
api_key	M	H		Valid API Key
JustpayCode	M	Q	A 30	Justpay code for APP
RetrievalReference	M	Q	N 12	Retrieval reference from transaction request or

Sample Request

```
https://dev.apigateway1.seylan.ad:2260/QR/LankaQR/1.0/InquireLankaQRTransaction?RetrievalReference=E210506180954$$AA0000008654715&JustpayCode=6289_M001_001
```

Sample Response

```
{
  "LankaQRTransactionStatus": {
    "Status": {
      "Transaction_Reference": "E220310121313$$AA0000009611122",
      "Timestamp": "2022-03-10T06:43:16.647Z",
      "Code": "0000",
      "Message": null,
      "Description": null
    },
    "Original_transaction_reference": "E220310121313$$AA0000009611121",
    "Transaction_date": "2022-03-10",
    "Transaction status": "SUCCESS",
    "Debit_RRN": null,
    "Debit_STAN": null,
    "Credit_RRN": null,
    "Credit_STAN": null
  }
}
```

Status code reference

Status Code	Type	Description
0000	S	Success
1010	E	Internal communication failure with remote host
2020	E	Request rejected by remote host. - Remote server returned failed status. Refer Description field for more information

6. Visa/Master QR Acquirer Services

6.1 Initiate Visa/Master QR Transaction [POST]

This service will initiate a LankaQR transaction using JustPay registered account

Request Parameters

Parameter	Req?	Type	Format	Description
api_key	M	H		Valid API Key
Device_id	M	B	A 8 - A 45	Device Id
Institution_id	M	B	N 3	User the fixed value "1"
Party_id	M	B	N 15	Unique id assigned to channel application. Channel will receive this credential from Bank or over auto-generated email
Function	M	B	A 30	Use the fixed value "ThirdPartyScanNPay"
Channel_user_id	M	B	A 100	Channel user Id
Channel_password	M	B	A 100	Channel Password
Card_type	M	B	A 1	Card Type Visa (V) / Master (M)
Retrieval_reference	M	B	A 30	Unique request reference number
Transaction_sequence	M	B	N 12	Transaction unique sequence
QR_payload	M	B	A 512	QR Payload
Customer_card_number	M	B	A 20	Customer account number
Customer_name	M	B	A 50	Customer name
Merchant_pan	M	B	A 28	Merchant Pan TAG 26 value need to be passed
Merchant_name	M	B	A 25	Merchant name
Transaction_ccy	M	B	A 3	Transaction currency code
Transaction_ccy_num	M	B	N 3	Transaction currency number
Transaction_amount	M	B	M 12	Transaction amount
Tip_indicator	O	B		Tip Indicator 01 – Tip Amount need to be passed 02 – Convenience fee 03 – Convenience fee percentage
Tip_amount	O	B	M 12	Tip Amount in case of Tip_indicator = '01'
Convenience_fee	O	B	M 12	Convenience fee amount or percentage in case of Tip_indicator is '02', '03'
Merchant_catg_code	M	B	A 4	Merchant category code
Merchant_city	M	B	A 25	Merchant city
Country_code	M	B	A 5	Merchant country code
Postal_code	M	B	N 6	Postal code
Transaction_desc	M	B	A 50	Transaction Description/Narration
Payment_ref1_code	O	B		If Tag62 is present in QR then this should be sent Possible values 01 - BILL_NUMBER 02 - MOBILE_NUMBER 03 - STORE_ID 04 - LOYALTY_NUMBER 05 - REFERENCE_ID

				06 - CONSUMER_ID 07 - TERMINAL_ID 08 – PURPOSE
Payment_ref1	O	B		If Tag62 is present in QR then this should be sent
Payment_ref2_code	O	B		If Tag62 is present in QR then this should be sent Possible values 01 - BILL_NUMBER 02 - MOBILE_NUMBER 03 - STORE_ID 04 - LOYALTY_NUMBER 05 - REFERENCE_ID 06 - CONSUMER_ID 07 - TERMINAL_ID 08 – PURPOSE
Payment_ref2	O	B		If Tag62 is present in QR then this should be sent
Transaction_date	M	B	D	Transaction date
Transaction_time	M	B	T	Transaction time
IP_address	O	B	A 50	IP Address
Session_id	O	B	A 50	Session Id
Check_value	M	B	A 500	Check value for the request. Calculation method is included below

Check value calculation

Note:

Data string used for the check value calculation should be constructed as below

Channel_user_id|Retrieval_reference|Function| | Payment_mode|Transaction amount|Merchant_name

Calculation method

```
import javax.crypto.Mac;
import javax.crypto.SecretKey;
import javax.crypto.spec.SecretKeySpec;
public class GenerateChecksum
{
    private final char[] HEX_TABLE = { '0', '1', '2', '3', '4', '5',
    '6', '7', '8', '9', 'A', 'B', 'C', 'D', 'E', 'F' };
    //secret Key - Key shared between channel and Euronet
    //Data String is pipe delimited values
    public String getSecureSHAHash(String secretKey, String dataString)
    throws Exception
    {
        byte[] mac = null;
        SecretKey key = new SecretKeySpec(secretKey.getBytes("UTF-8"), "HmacSHA512");
        Mac m = Mac.getInstance("HmacSHA512");
        m.init(key);
        m.update(dataString.getBytes("UTF-8"));
        mac = m.doFinal();
    }
}
```

```

        String hashValue = hex(mac);
        return hashValue;
    } //End of hex

    public String hex(byte[] input) throws Exception
    {
        StringBuffer sb = new StringBuffer(input.length * 2);
        for (int i = 0; i < input.length; i++)
        {
            sb.append(this.HEX_TABLE[(input[i] >> 4 & 0xF)]);
            sb.append(this.HEX_TABLE[(input[i] & 0xF)]);
        }
        return sb.toString();
    }
}

```

Finally, the calculated checksum should be encoded with Base64

Sample Request

```
https://<<sbk-gateway:port>>/QR/VMQR/1.0/InitiateVMQRTransaction
```

Request Payload

```

{
  "VMQRTransaction_Request": {
    "Device_id": "ABCDE123-AB12-A123-AB12-ABCDEF123456",
    "Institution_id": "1",
    "Party_id": "220590000000001",
    "Function": "ThirdPartyScanNPay",
    "Channel_user_id": "API_AC88732",
    "Channel_password": "t#N-fw5esdf",
    "Retrieval_reference": "202202210395",
    "Card_type": "V",
    "QR_payload":
"00020155020101021102164157241220007606031641572412200076065802LK04155
419022220007602632002816288000162880000622000760905923Religious
Establishment0515541902222000760801000060000006010Colombo
0261050020062070503***52048661530314464470002EN0123Religious
Establishment0210Colombo 026304D32C",
    "Transaction_sequence": "202202210395",
    "Customer_card_number": "?C77XHVFY21CR64X",
    "Customer_name": "TheCustomer",
    "Merchant_pan": "16288000162880000622",
    "Merchant_name": "The Shop",
    "Transaction_ccy": "LKR",
    "Transaction_ccy_num": "",
    "Transaction_amount": "150",
    "Tip_indicator": "",
    "Tip_amount": "",
    "Convenience_fee": "",
    "Merchant_catg_code": "4201",
    "Merchant_city": "",
    "Country_code": "",
    "Postal_code": "",
    "Transaction_desc": "TEST QR TRAN 202202210395",
    "Payment_refl_code": "03",
    "Payment_refl": "Test",
    "Payment_ref2_code": "05",
    "Payment_ref2": "07183972999",
    "Transaction_date": ""
  }
}

```

```

    "Transaction_time": "",
    "IP_address": "",
    "Session_id": "",
    "Check_value":
"MDZBNTEyNTFEQ0Q5MDcyQTQxM0RGQTg2NTMwQzMzMzRCNUNGNzA3RjQ4QjNGRTJEODExN
0RFNDA4QjFBMEVBNTA4RDA2MzkyNkU5RUNCNTQzOTU4NDBFNjJBNEUxRTlCRUNFMjExMDc
0MEI1NDlEMThBOTVCMENC RDkwRDgxRkI=",
    "Application_type": "",
    "Input_branch": "",
    "Input_user": "",
    "Workstation_id": "",
    "Posting_batch": ""
  }
}

```

Sample Response

```

{
  "VMQRTransaction_Response": {
    "Status": {
      "Transaction_Reference": "E220630163155$$AA0000008379285",
      "Timestamp": "2022-06-30T11:02:01.835Z",
      "Code": "0000",
      "Message": "",
      "Description": ""
    },
    "RRN": "218111006993",
    "STAN": null
  }
}

```

Status code reference

Status Code	Type	Description
0000	S	Success
1010	E	Internal communication failure with remote host
2020	E	Request rejected by remote host
2070	E	Transaction failed
2100	E	Account does not exists.
2150	E	No sufficient funds
2130	E	Restricted Account
3020	E	Issuer unavailable
3030	E	Request declined by issuer

6.2 Inquire Visa/Master QR Transaction Status [GET]

This service returns the status of previously processed transaction and this must be used to check the status if the transaction service did not respond to the request.

Request Parameters

Parameter	Req?	Type	Format	Description
api_key	M	H		Valid API Key
RetrievalReference	M	Q	N 12	Retrieval reference from transaction request or

Sample Request

```
https://<<sbk-gateway:port>>/QR/VMQR/1.0/InquireVMQRTransaction?
RetrievalReference=E210506180954$$AA0000008654715
```

Sample Response

```
{
  "VMQRTransactionStatus": {
    "Status": {
      "Transaction_Reference": "E220704163813$$AA0000003447180",
      "Timestamp": "2022-07-04T11:08:15.832Z",
      "Code": "2020",
      "Message": "Request rejected by remote host",
      "Description": "PSD0253 Original transaction failed"
    },
    "Original_transaction_reference":
    "E220630163155$$AA0000008379285",
    "Transaction_status": "FAILED",
    "Transaction_date": "2022-06-30",
    "RRN": null,
    "STAN": null
  }
}
```

Status code reference

Status Code	Type	Description
0000	S	Success
1010	E	Internal communication failure with remote host
2020	E	Request rejected by remote host. - Remote server returned failed status. Refer Description field for more information

7. QR Merchant API services

7.1 Generate QR [POST]

Note: Value marked with * will be provided by Bank during initiation of Integration. Kindly contact Bank in case details is not received. In case any parameter is not clear / need additional information, kindly contact Bank. “Parameter Name” are case sensitive, kindly ensure values used as same as provided below.

Request Parameters

Parameter	Req?	Type	Format	Description
api_key	M	H		Valid API Key
Institution_id	M	B	N1 – N3	Institution ID
Channel_user_id	M	B	A5 – A15	Channel user ID*
Channel_pass	M	B	A5 – A100	Channel Password*
Request_ref_no	M	B	N1 – N12	Unique request reference number
Merchant_login_id	M	B	A2 – A20	Merchant User ID
Merchant_login_pass	M	B	A5 – A20	Merchant User Password
Function	M	B	A8 – A20	generateQRAPI
Device_id	O	B	A0 – A50	User Device id which is used for device binding
Type	M	B	N1	“0” – For Static QR Generation, “1” – For Dynamic QR Generation
Mid	M	B	N1 – N15	Merchant ID
Tid	M	B	N1 – N10	Terminal ID
Transaction_amount	C	B	M15	Transaction Amount as per Merchant Currency
Transaction_currency	C	B	N3	Numeric code of Transaction Currency
Bill_no	O	B	A0 – A25	Customer Bill Number
Mobile_no	O	B	A0 – A25	Customer Mobile Number
Store_label	O	B	A0 – A25	Store Label
Loyalty_number	O	B	A0 – A25	Loyalty Number
Reference_label	O	B	A0 – A25	Reference Label
Customer_label	O	B	A0 – A25	Customer Label
Terminal_label	O	B	A0 – A25	Terminal Label

Purpose_of_transaction	O	B	A0 – A25	Purpose of Transaction
Additional_customer_data_request	O	B	A3	In case additional customer information. Allowed values: E: For Email, M: For Mobile, A: For Address Possible Values: E, A, M, EA, EM, MA, EMA etc (Any combination) based on requirement.
Session_id	O	B	A0 – A50	Session ID
Ip_address	O	B	A0 – A128	IP Address of source device
Check_sum	M	B	A0 – A500	calculated checkvalue/checksum

Checksum value calculation

Note:
Data string used for the check value calculation should be constructed as below
Request_ref_no|Mid|Tid|Merchant_login_id|Type|Transaction_amount|
Bill_no|Mobile_no

**** Please refer the Annexure 1 for further details of checksum value calculation**

Sample Request

<https://<<sbk-gateway:port>>/MerchantQR/1.0/GenerateQR>

Request Payload

```
{
  "GenerateQR_Request": {
    "Institution_id": "1",
    "Channel_user_id": "MerchantAPI",

"Channel_pass": "MzMwRjg0ODM3M0JENjQ1QTBDNkNCMDFEMTgyODg1RDk4QjNBN0UxODg1QTJCRTZzMkNCRTgzQzczMEE1MDg3Qg==",
    "Request_ref_no": "272006510084",
    "Merchant_login_id": "API_XGaSy82R74",
    "Merchant_login_pass": "tB2MeDDV95B5",
    "Function": "generateQRAPI",
    "Device_id": "51c7590c-6ce4-47db-a688-038bfd48bca0",
    "Type": "1",
    "Mid": "24235000000312",
    "Tid": "24000873",
    "Transaction_amount": "150",
    "Transaction_currency": "144",
    "Bill_no": "8867582",
    "Mobile_no": "7788998878",
    "Store_label": "",
    "Loyalty_number": ""
  }
}
```



```

        "Reference_label": "",
        "Customer_label": "",
        "Terminal_label": "",
        "Purpose_of_transaction": "",
        "Additional_customer_data_request": "EMA",
        "Session_id": "",
        "Ip_address": "",
        "Check_sum":
"MUFGMDNBNEU4NkFDNUJCN0MzMzFFMjk2MEE2NjRBQkVCQkREQjBDQUI2RUVENjE0NkE1Q
TJCQkU0NDclQTg4NzcXOTU0RDU5Rki4M0EyQkVFOTQzNjdERTgzOUQ1RkU2MUY2MjQ1MEM
0Mzc0RDhEREJBRTQ2Mzk0QjlDOUY4Q0U="
    }
}

```

Sample Response

```

{
  "GenerateQR_Response": {
    "Status": {
      "Transaction_Reference": "E240715142328$ISE0000006123103",
      "Timestamp": "2024-07-15T08:53:35.951Z",
      "Code": "0000",
      "Message": null,
      "Description": "EN00::Success"
    },
    "QR_Information": {
      "Institution_id": "1",
      "Channel_user_id": "MerchantAPI",
      "Interchange_list": [
        "V",
        "M",
        "L"
      ],
      "QR_code":
"
Hn01EQVR42u3d0XLbOgWE0Pz/T7fv7UynBhaQaJ59bBonlg494Iq99+eXyDX5cQkEdxHcR
XAXwV0EdxHcRXAXwV0EdxHcRXAX3EVwF8FdBHcR3EVwF8FdBHcR3EVwF8FdcBfBXQR3Edx
FcBfBXQR3EdxFcBd5DfefXj56wb//zkd/8j/vovmCzbdcu/LxF/zHe0/dZdxxxx133HF/I
ffmd/3PUJ79hvHVUlsSTUzN91770oIN3HHHXXfccX+ce234jo+/NXmlex/fnDTn6eY7Tb1
13HHHXXfccb+Ke0pnildqVzC3jONrtbmzwh133HHHXXeze+R2ppqyn3TiPV188TdXHe644
4477rh/DfeFqTflyguLLX7cKv5Utblrin8X7rjjjjvuUD/FvXnP/MkRf+K8O+6444477t/
H/ZFdQeqUldx+olkXLnyaPD5q44477rjjjvuJs3vzu+LV20sG2U2dtTXWHP1xxx133HHH/
QjutXcbXy3NX2NOeVxw8yh8c7OEO+6444477ndyXxjQ57q8VLPZ3EI010Tq10g1rbjjjjv
uuON+BPfNsXVhkE1VeHOFb/MWNMtc3HHHXXfccf8a7rUvle5Hs0CML7/m2ki9curjqfaBh
Tvuu0000+5HcI9r2DxlVZuMa0srXsvGe8xU9/rNT1Vxxx133HG/7Lx7U/DcSpjrFudG7dR
T1c1Hy9+8VcUdd9xxx/20/0Zkqr1aeFQ5V7DGtzS1xb/wTLd5NXDHXXfccf9PUVks4OLV
3ipbnFu+aVWXWqNpZ5544477rjjjvuh3Od6qNQxsvhkPDfxx1fCwpcuPUSAO+644477FxW
Rzelw89trU3h8eo7vLmo/K/75ctc/78Add9xxx/382X2hg1uoAlOUU+fCm2e8agXrm9tG3
HHHXXfccX92iI8ffF+YjJvblbmHxHPzdO3u3D6744477rjjfn4RGa/MamLia7U5K8efqm5
+V+0Fr3iqijvuu000+2VFZMp9ap9QG1tTYhTu4t4jxmvUy99qoo77rjjjvuxs3ttvp+bn
ms3r9bKNd9gfLHNduLrxbccccdd9xxfzP3+P1I/da4i+bYGl+9m095H+8fccdd9xxxz3
FPXXqK36t55rNeI8ZH6wXFsntwhwxx133HE/n3t8Okz1j82ltdBsphZbvN5Nbai++RAB7
rjjjjjvut/6vyOLzYrx2jG8hmgfUFuDG9ySXFpG444477rifzz01K6dOWW1Oq3Nbkdt2YO5
+pbY0u0000+644/6eInLuqeFcC/bRIImnSaY7sc19KvS/ccccdd9xxP72ITNVqKZ0L9Vxz2
cxhmrtQzTNnu0000+644/7C2T0uuNaUNbcQR51xmwspctc1ofszhjjvuu000+lnca/Nis++
r/fT4qmsuyKbgeHnanO9xxx133HHH/Ygicq7H/OgGxyfj2k9PXai5x7219RO/8rjjjjvuu
ON+xOyeai3j5VccU+1ceHx2r73B5u8c/yjEHXfcccd96e4P9v1zU3G8ao0vu3ZrHdTWxr
ccccdd9xxfw/32jWqNYC1XmwBbvwha00vz33JIQLccccdd9xv4N5s5VLN5twQv9m9vo37w
p4Ed9xxxx133Je5L5zVjh/abq6f2luOv/f4yfXaXH5XEYk77rjjjvut/52ZVCOZup1z9yP
l/qPrM/f4eeGfAeCOO+644477odxTp8mbleIjS3STRXmLpOZ73HHHXXfccT+riGxe63ht1

```

```

1yrqYE4fuJ84WBZ87t+7QZ33HHHXXfc57g3r+wj/4Ss9humnlA2r0/zh9b+stkd9xxxx3
304vIprPmme/NwXruvNTm3ib1SZE6x4877rjjjjvuj3OfIzhXKab2Cc2Ze47p3As6RIA77
rjjjvtVReRCyfjIU9XmkBo/TT7X2M6N/rjjjjvuuOP+xbP75oqK3+D48bjmyqwVmnO1LO6
444477ri/cHZvto3N0XZzgow3pPGxPnURPFXFHxfcccf9qtm9Nmo3X6dZKc7Vl3NdXqq+T
E38jyvHHXfccccd94UiMlWrNf9JW7ziTM24zck4VvammOKOO+644477983uTcG1I2JzKJs
NaXOFNzc5cwf6H3ePO+6444477nHcc71Y84IuHNuam93jHyvxxY877rjjjjvuX8M9fmS8e
T/mnqo+smibi6S5bOZ6TNxxxx133HF/vIhc2ANsHvZaaAkXDpbVPmhS7Sfu0000+64nz6
7L0x+tbu4cOorNSsv3J3Uu7i9iMQdd9xxx/1Y7nNN2UcX/ZH9Rvydzq2Nhd9xxxx133
E/n3pwOm/djob6M71LiHxnNPvTZ94U77rjjjjvup3OvFYjPPlVtDtbN0jM1ajfvqdkdd9x
xxx133IdKvfhzzfgD4NrxuLnVm6ovcccd9xxx/1Q7qlb3lw/HxWic6XnwiKZay3jHzS44
4477rjj/mbuzw7EzbElvoVoloOpBZna5DQl4I477rjjjvubuYu8JLg7iK4i+AugrsI7iK
4i+AugrsI7iK4i+AuuiVgLoK7CO4iuiVgLoK7CO4iuiVgLoK74C6CuwjuIriL4C6CuwjuI
riL4C6Cu8gf+Q3QcNfc2/WGeQAAAABJRu5ErkJggg==",
    "Request_ref_no": "272006510040",
    "Response_code": "EN00",
    "Response_description": "Success",
    "Check_value":
    "Rjg0MDk4MEU3Rjg2NzQzREY0OTZDNuY1Nzc5REUzM0ZCODRBNjM5REMxRkM1MTE0OTkxN
DhGRjAwOERGNzg3NUJDN0I4NzQ5N0E0OUE3MzAyOERGOERENjLGRDY0MjZfQTc2MDE3NzU
0MjU5REFDQjY3QkIzMkRGNTZCNTFGM0M=",
    "Bank_name": "Seylan Bank",
    "Bank_name_flag": "Y",
    "Bank_logo_flag": "Y"
  }
}
}

```

Status code reference

Status Code	Type	Description
0000	S	Success
1010	E	Internal communication failure with remote host
2010	E	Record not available. Invalid search key.
2030	E	Account validation failed
2060	E	Request Exceeds the maximum range allowed

7.2 QR Transaction Inquiry [POST]

If merchant is passing Customer Bill Number, Mobile Number, Customer ID etc. fields in dynamic QR under Additional Tag 1 and Additional Tag 2, then this information / values can be searched based using Additional_value_first and Additional_value_second parameters mentioned above.

Merchant can pass value that needs to be searched in either of the field and basis input QR Engine will check for transactions matching with these values and will give appropriate response.

If merchant is providing values in Additional_value_first or Additional_value_second, then From_date and To_date values are mandatory, system will reject request if values for From_date and To_date as not provided in request.

Request Parameters

Parameter	Req?	Type	Format	Description
api_key	M	H		Valid API Key
Institution_id	M	B	N1 – N3	Institution ID
Channel_user_id	M	B	A5 – A15	Channel user ID*
Channel_pass	M	B	A5 – A100	Channel Password*
Request_ref_no	M	B	N1 – N12	Unique request reference number
Merchant_login_id	M	B	A2 – A20	Merchant User ID
Merchant_login_pass	M	B	A5 – A20	Merchant User Password
Function	M	B	A8 – A20	viewMerTxnAPI
From_date	C	B	A10 – A10	From date (YYYY-MM-DD)
To_date	C	B	A10 – A10	To date (YYYY-MM-DD)
Rrn	C	B	A10 – A20	Either only retrieval reference no based search
Mid	M	B	N1 – N15	Merchant ID
Tid	M	B	N1 – N10	Terminal ID
Customer_card_no	C	B	N14 – N19	Either only customer card number-based search
Additional_value_first	C	B	A0 – A25	Additional Value Alphanumeric with Special Characters (<>';`&!~ not allowed)
Additional_value_second	C	B	A0 – A25	Additional Value Alphanumeric with Special Characters (<>';`&!~ not allowed)
Records_per_page_count	C	B	N1 – N3	Number of Records to display in a page
Page_no	C	B	N1 – N2	Number of Page to display in a page
Ip_address	O	B	A0 – A128	IP Address of source device
Check_sum	M	B	A0 – A500	calculated checkvalue/checksum

Checksum value calculation

Note:

Data string used for the check value calculation should be constructed as below

`Merchant_login_id|From_date|To_date|Mid|Tid|Rrn`

**** Please refer the Annexure 1 for further details of checksum value calculation**

Sample Request

`https://<<sbk-gateway:port>>/MerchantQR/1.0/TransactionView`

Request Payload

```
{ "TransactionView_Request": {
    "Institution_id": "1",
    "Channel_user_id": "MerchantAPI",
    "Channel_pass": "MzMwRjg0ODM3M0JENjQlQTBDNkNCMDFFEMTgyODg1RDk4QjNBN0UxODg1QTJCRTEzMkNCRTgzQzczMEE1MDg3Qg==",
    "Request_ref_no": "272007210070",
    "Merchant_login_id": "API_QR81493",
    "Merchant_login_pass": "6ehi-UPG+F",
    "Function": "viewMerTxnAPI",
    "From_date": "2024-06-01",
    "To_date": "2024-06-22",
    "Rrn": "",
    "Mid": "242000000000298",
    "Tid": "24000860",
    "Customer_card_no": "",
    "Additional_value_first": "",
    "Additional_value_second": "",
    "Records_per_page_count": "10",
    "Page_no": "1",
    "Ip_address": "192.131.1.1",

    "Check_sum": "RTJFMj1DNTA1MkY3MjM4OTJEQzEzNDVCRkE0NUIwNkI3NTE2M0JFRtMzNzVGRUZfQTIxNzU3N0I5Qzg5NjYyMjIzQjNGOTAYNDRDQTk3RTYxQ0Q1RkUxM0IwQjQ4NzI1RTY0N0RERTkxMDA1MjUzQzIwMDFFODkwRjQwRjJBQzE="
  }
}
```

Sample Response

```
{
  "TransactionView_Response": {
    "Status": {
      "Transaction_Reference": "E240709200600$ISE0000006357981",
      "Timestamp": "2024-07-09T14:36:40.738Z",
      "Code": "0000",
      "Message": null,
      "Description": "EN00::Success"
    },
    "TransactionView_Information": {
      "Institution_id": "1",
      "Channel_user_id": "MerchantAPI",
      "Request_ref_no": "272007210067",
    }
  }
}
```

```

"Total_records": "2",
"transactionDetails": [
    {
        "Retrieval_refernce_no": "417403004474",
        "Stan": "004474",
        "Merchant_name": "QR API Merchant 1",
        "Mid": "24164000000237",
        "Tid": "24000903",
        "Merchant_pan": "162880-XXX-XXX-0624",
        "Merchant_category_code": "5399",
        "Merchant_location": "Colombo 03",
        "Tip_amt": "0.00",
        "Transaction_amount": "3.00",
        "Transaction_fee_amtount": null,
        "Transaction_currency": "144",
        "Transaction_date": "2024-06-22 09:12:11.92",
        "Transaction_description": "Third Party(Refund)",
        "Interchange": "S",
        "Card_no": null,
        "Customer_account_no": null,
        "Response_code": "EN76",
        "Response_description": "FULL REVERSAL",
        "Auth_code": "000483",
        "Additional_tag1_name": null,
        "Additional_data1": null,
        "Additional_tag2_name": null,
        "Additional_data2": null,
        "Transaction_hash_value": null,
        "Scheme_description": "Onus"
    },
    {
        "Retrieval_refernce_no": "417307004473",
        "Stan": "004473",
        "Merchant_name": "QR API Merchant 1",
        "Mid": "24164000000237",
        "Tid": "24000903",
        "Merchant_pan": "162880-XXX-XXX-0624",
        "Merchant_category_code": "5399",
        "Merchant_location": "Colombo 03",
        "Tip_amt": "0.00",
        "Transaction_amount": "250.00",
        "Transaction_fee_amtount": null,
        "Transaction_currency": "144",
        "Transaction_date": "2024-06-21 12:43:52.977",
        "Transaction_description": "Third Party(Debit)",
        "Interchange": "S",
        "Card_no": null,
        "Customer_account_no": "001201-XXX-XXX-3100",
        "Response_code": "EN00",
        "Response_description": "Success",
        "Auth_code": "000483",
        "Additional_tag1_name": "05",
        "Additional_data1": "0503s.1",
        "Additional_tag2_name": null,
        "Additional_data2": "null",
        "Transaction_hash_value": null,
        "Scheme_description": "Onus"
    }
],
"Response_code": "EN00",
"Respose_description": "Success",

```

```

    "Check_value":
    "NkZFNDE3RDMwQTlFRUQ4OEUwRkQ5NTA3QjU3RjcwNTg0OTJFMzNCQUUwRjgyMTlCNDgzQ
    ThDOTRCMTEyNkE4OEFDMEU5OTBCQjg5QzBBODhDQkVGRjhBM0ZFNUQ2QzQzRDBEM0IyMDF
    BRTQ1Q0Q1NEZFQjQ2QTUxMDFFMdNCNzc="
  }
}
}

```

Status code reference

Status Code	Type	Description
0000	S	Success
1010	E	Internal communication failure with remote host
2010	E	Record not available. Invalid search key.
2030	E	Account validation failed
2060	E	Request Exceeds the maximum range allowed

7.3 QR Transaction Refund [POST]

Merchant **MUST** ensure refund is not processed multiple times for same transaction in a day. In case of no response or time out response merchant should not initiate refund again and should contact bank for support for refund status.

Request Parameters

Parameter	Req?	Type	Format	Description
api_key	M	H		Valid API Key
Institution_id	M	B	N1 – N3	Institution ID
Channel_user_id	M	B	A5 – A15	Channel user ID*
Channel_pass	M	B	A5 – A100	Channel Password*
Request_ref_no	M	B	N1 – N12	Unique request reference number
Merchant_login_id	M	B	A2 – A20	Merchant User ID
Merchant_login_pass	M	B	A5 – A20	Merchant User Password
Function	M	B	A8 – A20	merRefundAPI
Stan	O	B	A1 – A10	Merchant Transaction STAN
Retrieval_refernce_no	M	B	A12 – A12	Merchant Transaction retrieval reference no
Refund_amount	M	B	A1 – A15	Amount that merchant wishes to refund to customer
Interchange	M	B	A1 – A1	Interchange used for transaction, like Visa/MC/Lanka Pay/ONUS

Ip_address	O	B	A0 – A128	IP Address of source device
Check_sum	M	B	A0 – A500	calculated checkvalue/checksum

Checksum value calculation

Note:

Data string used for the check value calculation should be constructed as below

Merchant_login_id|Retrieval_refernce_no|Interchange

**** Please refer the Annexure 1 for further details of checksum value calculation**

Sample Request

<https://<<sbk-gateway:port>>/MerchantQR/1.0/MerchantRefund>

Request Payload

```
{ "MerchantRefund_Request": {
  "Institution_id": "1",
  "Channel_user_id": "MerchantAPI",

  "Channel_pass": "MzMwRjg0ODM3M0JENjQ1QTBDNkNCMDFEMTgyODg1RDk4QjNBN0UxODg1QTJCRTEzMkNCRTgzQzczMEE1MDg3Qg==",
  "Request_ref_no": "282007210063",
  "Merchant_login_id": "API_QR77621",
  "Merchant_login_pass": "4WC*67CSujB) ",
  "Function": "merRefundAPI",
  "Stan": "004473",
  "Retrieval_refernce_no": "417307004473",
  "Refund_amount": "4",
  "Interchange": "S",
  "Ip_address": "1.1.1.1",

  "Check_sum": "Q0MzM0I0NDZGMTRDRUJBQkNBOUYwMDI1MjYyM0E3NUVEOTBBNUU1NjRDOEU5RjcyMEQxQzFEODE5RUQxMEJCNUEzNzA2NTU0Qz1FREUzNDBBQjIyQzkyNkFBRDA4MjgzMEFBRkExNTA3REYyNEYzRjg5RTE5MkrDRUQ1OT1BOEM="
}
}
```

Sample Response

```
{
  "MerchantRefund_Response": {
    "Status": {
      "Transaction_Reference": "E240709204752$ISE0000005153691",
      "Timestamp": "2024-07-09T15:18:16.697Z",
      "Code": "2020",
      "Message": "Request rejected by remote host",
      "Description": "EN20::Duplicate ReqRefNo"
    },
    "MerchantRefund_Information": {
      "Institution_id": "1",
      "Channel_user_id": "MerchantAPI",
      "Request_ref_no": "282006210063",

```

```

    "Response_code": "EN20",
    "Response_description": "Duplicate ReqRefNo",
    "Check_value": null
  }
}
}

```

Status code reference

Status Code	Type	Description
0000	S	Success
1010	E	Internal communication failure with remote host
2010	E	Record not available. Invalid search key.
2030	E	Account validation failed
2060	E	Request Exceeds the maximum range allowed

Annexure 1

Checksum Generation Method

To generate checksum, checksum key will be provided by the bank before start the integration testing. This key MUST be stored securely with limited access to only required authorized personnel. This key will be used to generate checksum for transaction request and response. Request and response should be acknowledged and processed only post verification of checksum values. Matching of checksum will ensure there is no man in middle attack while data flows between channel/partner and QR Engine.

Checksum needs to be created using HMAC-SHA512 algorithm. Checksum generation highly depends on the order or sequence of parameters set, hence its important order of parameters to be followed as decided. Checksum will be further encoded with Base64 encoding before passing in request or response.

Important Notes:

- a) Checksum Key will be different for UAT and Production environment.
- b) Calculated checksum should be encoded with base64 before passing into message.

Sample code to generate checksum:

Below code generates checksum by calling getSecureSHAHash method.

```
import javax.crypto.Mac;
import javax.crypto.SecretKey;
import javax.crypto.spec.SecretKeySpec;
public class GenerateChecksum
{
    private final char[] HEX_TABLE = { '0', '1', '2', '3', '4', '5', '6',
    '7', '8', '9', 'A', 'B', 'C', 'D', 'E', 'F' };

    //secret Key - Key shared between channel and Euronet
    //Data String is pipe delimited values
    public String getSecureSHAHash(String secretKey, String dataString)
    throws Exception {
        byte[] mac = null;
        SecretKey key = new SecretKeySpec(secretKey.getBytes("UTF-8"),
        "HmacSHA512");
        Mac m = Mac.getInstance("HmacSHA512");
        m.init(key);
        m.update(dataString.getBytes("UTF-8"));
        mac = m.doFinal();
        String hashValue = hex(mac);
        return hashValue;
    }

    public String hex(byte[] input) throws Exception{
        StringBuffer sb = new StringBuffer(input.length * 2);
        for (int i = 0; i < input.length; i++) {
            sb.append(this.HEX_TABLE[(input[i] >> 4 & 0xF)]);
            sb.append(this.HEX_TABLE[(input[i] & 0xF)]);
        }
        return sb.toString();
    }
}
```

Note: Checksum returned by above program should be base64 encoded before sending to EN API. Similarly, response from EN API will have base64-encoded checksum.

All parameters mentioned below for checksum generation are required to send in request with actual length and in define sequence only. If any tag has blank/NULL/null value or Merchant is not providing any tag which are required for checksum generation then Merchant should add blank value (Not a space) for that tag while creating checksum.

For example:

QR generation request tags consider for checksum: -

Request_ref_no|Mid|Tid|Merchant_login_id|Type|Transaction_amount|Bill_no|Mobile_no

- a. If Tid tag value is blank or tag is missing, then Merchant should generate checksum using below logic.

Request_ref_no|Mid| |Merchant_login_id|Type|Transaction_amount|Bill_no|Mobile_no

- b. If Mobile_no tag value is blank or tag is missing, then Merchant should generate checksum using below logic.

Request_ref_no|Mid|Tid|Merchant_login_id|Type|Transaction_amount|Bill_no|

Parameters for checksum generation

1. Generate QR

Request parameters for checksum generation

Request_ref_no|Mid|Tid|Merchant_login_id|Type|Transaction_amount|Bill_no|Mobile_no

Response Parameters for checksum generation

Request_ref_no|Response_code

2. View Transaction

Request parameters for checksum generation

Merchant_login_id|From_date|To_date|Mid|Tid|Rrn

Response Parameters for checksum generation

Request_ref_no|Response_code

3. Merchant Refund

Request parameters for checksum generation

Merchant_login_id|Retrieval_refernce_no|Interchange

Response Parameters for checksum generation

Request_ref_no|Response_code

Merchants can try using below sample for checking checksum value generated by merchant is matching with sample values provided. Sample below is for Generate QR

Parameter Name	Sample Value Used
Request_ref_no	272006210017
Mid	23085000000112
Tid	23000383
Merchant_login_id	API_testAPI57390
Type	1
Transaction_amount	10
Bill_no	88675777
Mobile_no	7788998877

Input String:

272006210017|23085000000112|23000383|API_testAPI57390|1|10|88675777|7788998877

Checksum Key Used for Sample Value Generation:

ar2t3nqh185wiro8zq0wzpcana6010ju

Checksum Generated:

QTMxNEEyMThGODUwMTAxOTFEQ0Q2RTI2QkE5NkZGRDhBQkQ1MjYyQUY0OUEzQ0VEOUEyQkl5OD
A3RkREMTU0ODgwN0E3ODUyM0Q5Q0ZFMTRFNTRFNEZDNjg3N0EzMjl4NzdCNTMzQzI5QjJBOUU0O
DRCRDizQzE3NjY4REUwRjQ=